Week 6 — AI Agent Integration with Google Antigravity IDE

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Project: WeatherTwin AI

Summary

The objective of this phase was to extend the WeatherTwin capstone project by integrating an AI agent layer using the Google Antigravity IDE. This transformation upgrades the existing weather dashboard into an intelligent decision-support application capable of autonomous reasoning, tool selection, and multi-step workflow execution.

Agent Tools Implementation

The following tools were developed in agent/tools.py to provide the agent with functional capabilities. All tools follow a standardized JSON response format for seamless LLM processing.

#	Tool	Functionality
1	get_current_weather	Fetches real-time weather via WeatherAPI.
2	get_weather_forecast	Returns hourly forecasts for a specified duration (N hours).
3	get_historical_analysis	Performs statistical analysis on local CSV datasets.
4	predict_weather_bert	Provides BERT-based classification for weather conditions.
5	compare_cities	Conducts side-by-side weather comparisons between two locations.

Technical Specifications

Agent Runner:

- Pattern: ReAct (Reason + Act).
- LLM: Groq API — llama-3.3-70b-versatile.

- Constraints: Maximum of 5 steps per query to ensure efficiency.
- Output: Generates a final answer accompanied by a detailed "Reasoning Trace" (Thought → Action → Observation).

Application Modes:

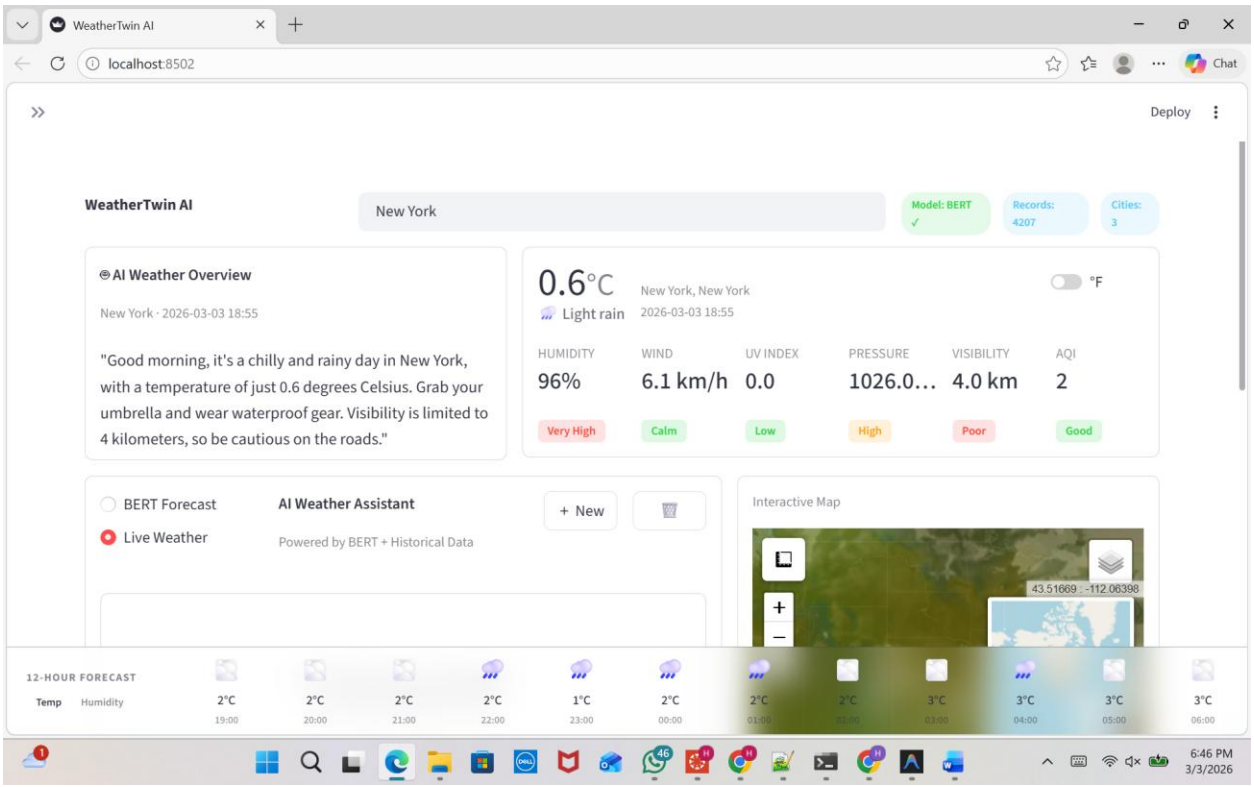
The Streamlit interface supports two distinct operational modes:

1. BERT Forecast Mode: Direct BERT classification on historical data without agent intervention.
2. Live Weather Mode: Full agent activation with multi-step tool usage and visible reasoning traces.

User Interface & Experience

The integrated dashboard features several key UI components designed for clarity and interactivity:

- Chat Interface: Clean user/AI bubble dialogue.
- Agent Trace Expander: A transparency feature allowing users to view the AI's step-by-step logic.
- Quick Action Chips: Pre-set queries for common activities (e.g., Beach Day, Hiking Trip, Ski Conditions).
- Visualisations: Interactive weather maps with city markers and a 12-hour forecast dock.

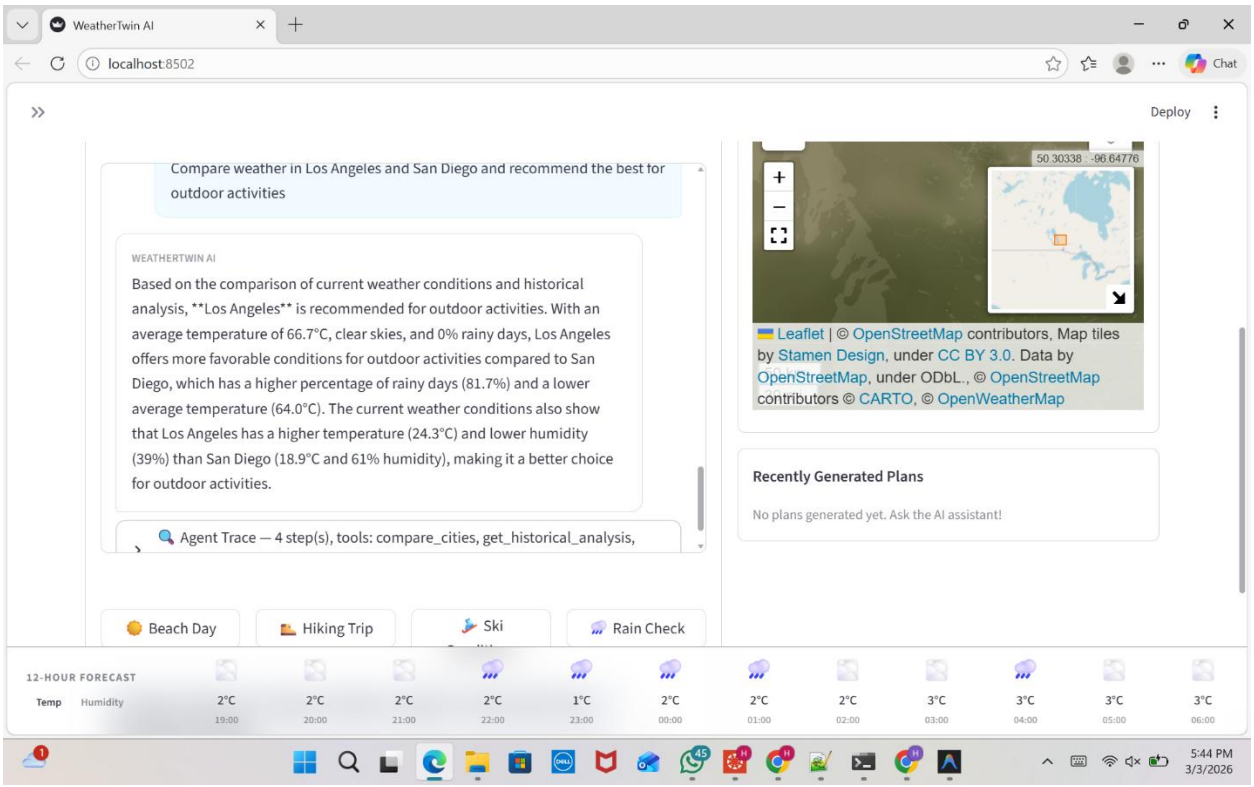


Interaction Analysis

The following logs demonstrate the agent's ability to handle varying levels of query complexity:

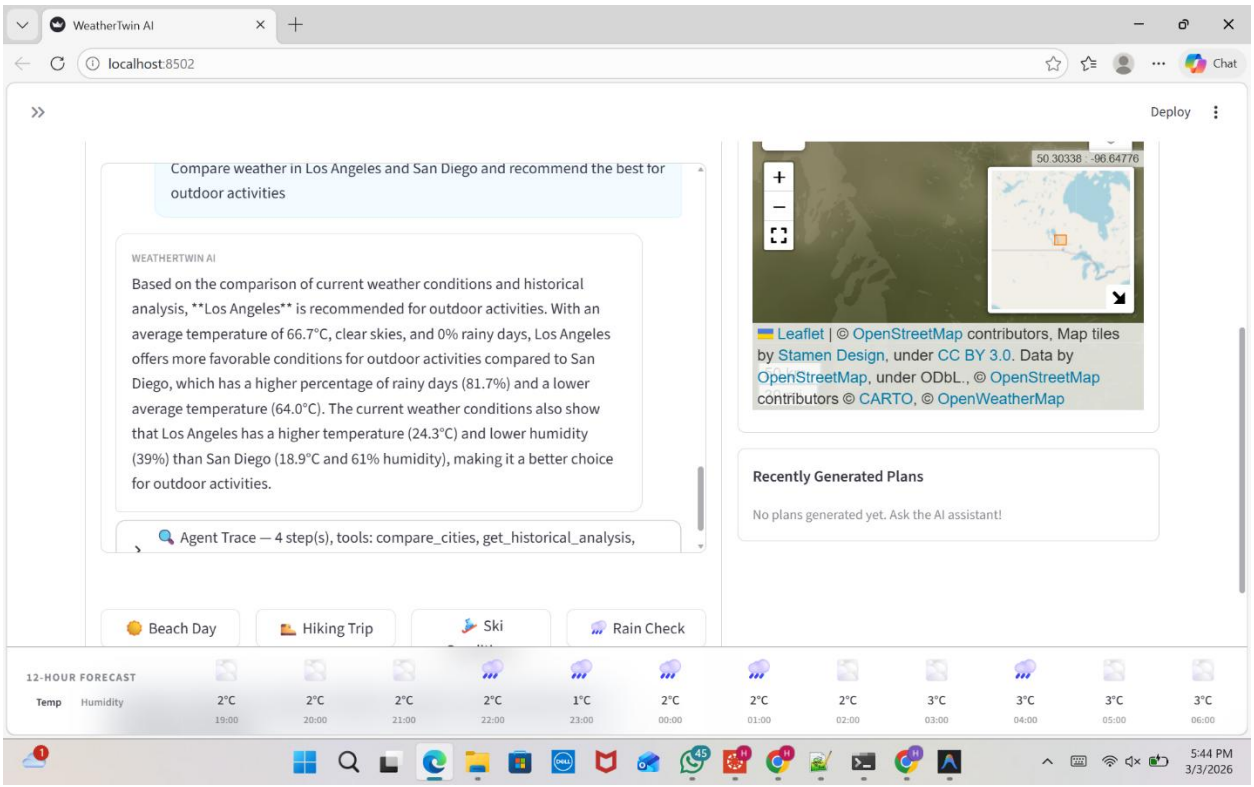
Multi-Step Reasoning (Activity Recommendation):

- Query: "Can I go hiking in San Francisco this weekend?"
- Tools Used: get_current_weather, predict_weather_bert.
- Outcome: The agent identified a "Negative" weather prediction (90.8% confidence) and advised the user to postpone the trip based on data rather than just current conditions.



Complex Comparison:

- Query: Comparing LA vs. San Diego for outdoor activities.
- Tools Used: compare_cities (triggering 4 nested tool calls).
- Outcome: Provided a data-driven recommendation for Los Angeles based on a significantly lower percentage of rainy days compared to San Diego's historical data (81.7%).



7. Evaluation & Performance

Scenario	Complexity	Response Time	Quality Assessment
Current Weather	Simple	~3s	Accurate; real-time.
Recommendations	Medium	~5s	Contextual; data-grounded.
Multi-City Comparison	Complex	~6s	Comprehensive; high utility.

Strengths:

- Grounds answers in real-time data from WeatherAPI.
- The Reasoning Trace serves as a built-in debugging and trust-building tool.
- Modular tool design allows for easy addition of new features.

Limitations & Future Work:

- Model Refinement: The BERT model currently uses a sentiment-based classifier; moving to weather-specific labels would improve nuance.
- Data Expansion: Historical data is currently limited to three California cities.
- Units: Occasional inconsistency in temperature units (°F vs °C).
- Optimization: Implementing caching for intermediate tool results to reduce API latency.

Setup & Technology Stack

- Frontend: Streamlit, Folium (Maps), Plotly.
- AI/ML: Groq (Llama 3.3), BERT (distilbert-base-uncased).
- Data Sources: WeatherAPI.com, OpenWeatherMap, Local CSVs.

Github link: https://github.com/Harsha-Sri24/DataScienceCapstone/tree/main/Week-6

Demo Video link:
https://drive.google.com/file/d/1ZIxZJZBqhN9NKkUsSSOrXI8CdceK8EMX/view?usp=sharing